Gravitational Coupling Control via Arithmetic Spectral Modulation: Gravity Nullification and Sign Reversal from the Euler Product Structure of Vacuum Energy

Wright Brothers[1]

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We prove two theorems within the spectral action framework applied to the Bost-Connes (BC) C*-dynamical system. **Theorem 1** (Gravity nullification): removing $K \geq 2$ prime Euler factors from the BC partition function $\zeta(\beta)$ creates a K-fold zero at $\beta = 0$, forcing the Seeley-DeWitt coefficients $a_0 = a_2 = 0$ and hence the effective gravitational coupling $G_{\text{eff}} = 0$. **Theorem 2** (Gravity reversal): π-phase rotation of all Euler factors replaces $\zeta(t)$ with the Liouville heat kernel $\zeta(t)/\zeta(2t)$, reversing the sign of a_2 and yielding $G_{\text{eff}} = -2G$. Both operations are implementable, in principle, via frequency-selective quantum elements (transmon qubits at $\omega_p = \log p/\tau$, in the standard microwave regime). We show that the $G = 0$ interior preserves the speed of light ($c_{\text{eff}} = c$, proved via the Connes distance formula), while the $G = -2G$ shell undergoes spectral dimension reduction to $d_S \approx 2$ (holographic). The framework predicts a falsifiable signature: the Casimir energy ratio in a 2-prime-sieved acoustic cavity equals 182 (Euler product) versus 1/3 (mode counting), testable with existing technology.

I. INTRODUCTION

The gravitational coupling constant G is conventionally treated as a fixed parameter of nature. In the spectral action principle [**?** **?**], however, G emerges from the Seeley-DeWitt coefficient a_2 of the Dirac operator D:

$$\frac{1}{16\pi G_{\text{eff}}} = f_2 \Lambda^2 \cdot \frac{1}{16\pi^2} \int_M \frac{R}{6} \sqrt{g}\, d^4x \Big/ \int_M R \sqrt{g}\, d^4x, \tag{1}$$

where $f_2 = \int_0^\infty f(u)\, du$ is the second moment of the cut-off function and Λ is the energy scale. If a_2 can be set to zero—or its sign reversed—then G is nullified or reversed.

The Bost-Connes (BC) C*-dynamical system [**?**] provides a natural arena for this possibility. Its partition function is the Riemann zeta function,

$$Z(\beta) = \text{Tr}(e^{-\beta H_{\text{BC}}}) = \zeta(\beta), \tag{2}$$

which admits a factorization into independent prime channels via the Euler product: $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$. Each prime p contributes an independent sector to the vacuum.

In this paper we show that removing or phase-rotating individual prime sectors gives precise control over G_{eff}.

II. THE BC HEAT KERNEL AND SEELEY-DEWITT COEFFICIENTS

Identifying $D^2 = H_{\text{BC}}$ (eigenvalues $\lambda_n = \log n$), the heat kernel is

$$K(t) = \text{Tr}(e^{-tD^2}) = \sum_{n=1}^{\infty} n^{-t} = \zeta(t). \tag{3}$$

The Seeley-DeWitt expansion for a d-dimensional geometry reads $K(t) \sim \sum_{k=0}^{\infty} a_{2k}\, t^{(2k-d)/2}$ as $t \to 0^+$. For $d = 4$:

$$K(t) = a_0\, t^{-2} + a_2\, t^{-1} + a_4 + a_6\, t + \cdots \tag{4}$$

The BC heat kernel $\zeta(t)$ is analytic at $t = 0$ with $\zeta(0) = -1/2$ and $\zeta'(0) = -\frac{1}{2}\log(2\pi)$. Matching to (4), the leading BC "Seeley-DeWitt coefficients" are encoded in the Taylor expansion at $t = 0$.

III. THEOREM 1: GRAVITY NULLIFICATION

Definition 1 (Prime-muted heat kernel)**.** *For a set of primes $S = \{p_1, \ldots, p_K\}$, define*

$$K_{\neg S}(t) = \prod_{p \in S} (1 - p^{-t})\, \zeta(t). \tag{5}$$

Theorem 2 (Gravity nullification)**.** *For $|S| = K \geq 2$, the muted heat kernel has a K-fold zero at $t = 0$:*

$$K_{\neg S}(t) = t^K \cdot \frac{(-1)^K}{2} \prod_{p \in S} \log p + O(t^{K+1}). \tag{6}$$

Consequently, in the Seeley-DeWitt expansion (4):

$$a_0 = a_2 = \cdots = a_{2(K-1)} = 0. \tag{7}$$

In particular, for $K \geq 2$: $a_0 = a_2 = 0$, so that the cosmological term and the Einstein-Hilbert term both vanish, giving $G_{\text{eff}} = 0$.

Proof. Each factor $(1 - p^{-t})$ has a simple zero at $t = 0$:

$$1 - p^{-t} = 1 - e^{-t \log p} = t \log p + O(t^2). \tag{8}$$

Therefore

$$\prod_{p \in S} (1 - p^{-t}) = t^K \prod_{p \in S} \log p + O(t^{K+1}). \tag{9}$$

Since $\zeta(0) = -1/2$ is finite and nonzero, we have

$$K_{\neg S}(t) = t^K \cdot \Big(-\tfrac{1}{2}\prod_{p\in S}\log p\Big) + O(t^{K+1}). \tag{10}$$

Comparing with $K(t) = a_0 t^{-2} + a_2 t^{-1} + a_4 + a_6 t + \cdots$, a function vanishing as t^K at the origin requires all coefficients of $t^{-2}, t^{-1}, \ldots, t^{K-3}$ to vanish: $a_0 = a_2 = \cdots = a_{2(K-1)} = 0$.

For $K = 2$: $a_0 = a_2 = 0$. No cosmological constant; no Einstein-Hilbert term; $G_{\text{eff}} = 0$. □

Corollary 3. *For $K \geq 3$, additionally $a_4 = 0$: the Gauss-Bonnet / Weyl-squared terms vanish. The leading gravitational interaction is of order $a_{2K} \neq 0$, a higher-derivative correction. For $K = 3$ (muting $\{2, 3, 5\}$): the geometry is locally flat to all orders up to 6th-derivative gravity.*

Physical content. The K-fold zero arises because each muted prime removes one "channel" from the Euler product, and at $t = 0$ (the UV point), this removal is maximally destructive. The gravitational coupling $G \propto a_2$ is the sum of contributions from all prime channels; removing two channels creates a cancellation that zeros a_2 exactly.

IV. THEOREM 2: GRAVITY REVERSAL VIA LIOUVILLE ROTATION

Definition 4 (Liouville heat kernel)**.** *The all-prime π-rotation replaces each Euler factor $(1-p^{-t})^{-1}$ with $(1+p^{-t})^{-1}$, giving*

$$K_\pi(t) = \prod_p (1+p^{-t})^{-1} = \frac{\zeta(2t)}{\zeta(t)}, \tag{11}$$

where the identity follows from $\zeta(t)/\zeta(2t) = \prod_p(1+p^{-t})$.

The Liouville heat kernel is connected to the Liouville function $\lambda(n) = (-1)^{\Omega(n)}$: the Dirichlet series $\sum_n \lambda(n) n^{-t} = \zeta(2t)/\zeta(t)$.

Theorem 5 (Gravity reversal)**.** *The Liouville heat kernel satisfies:*

1. *$K_\pi(0) = 1$ (no zero at $t = 0$).*
2. *$K'_\pi(0) = \log(2\pi)$.*
3. *The ratio $G^{(\pi)}_{\text{eff}}/G = K'_\pi(0)/\zeta'(0) = -2$.*

The effective gravitational coupling reverses sign: $G_{\text{eff}} = -2G$, corresponding to repulsive gravity.

Proof. (1) $K_\pi(0) = \zeta(0)/\zeta(0) = 1$.
(2) Writing $K_\pi(t) = \zeta(t)/\zeta(2t)$ and differentiating:

$$K'_\pi(t) = \frac{\zeta'(t)\zeta(2t) - 2\zeta(t)\zeta'(2t)}{\zeta(2t)^2}. \tag{12}$$

At $t = 0$: $\zeta(0) = -1/2$, $\zeta'(0) = -\frac{1}{2}\log(2\pi)$, and $\zeta(0) = \zeta(2\cdot 0)$, so

$$K'_\pi(0) = \frac{\zeta'(0)\cdot\zeta(0) - 2\zeta(0)\cdot\zeta'(0)}{\zeta(0)^2} = \frac{-\zeta'(0)}{\zeta(0)} = \log(2\pi). \tag{13}$$

(3) The Einstein-Hilbert coefficient is proportional to $K'(0)$. Normally, $K'(0) = \zeta'(0) = -\frac{1}{2}\log(2\pi)$. For the Liouville case:

$$\frac{G^{(\pi)}_{\text{eff}}}{G} = \frac{K'_\pi(0)}{\zeta'(0)} = \frac{\log(2\pi)}{-\frac{1}{2}\log(2\pi)} = -2. \tag{14}$$

□

Physical content. The π-rotation assigns a *fermionic* sign $(-1)^{\Omega(n)}$ to each mode. This is equivalent to replacing the bosonic vacuum with a fermionic one. The gravitational coupling reverses because fermions and bosons contribute oppositely to a_2 (a well-known feature of spectral geometry [**?**]). The factor of 2 arises from the ratio $\zeta(2t)/\zeta(t)$ having slope $\log(2\pi)$ versus the original slope $-\frac{1}{2}\log(2\pi)$ at $t = 0$.

V. LIGHT SPEED IN THE $G = 0$ REGION

Proposition 6 (Light speed invariance)**.** *If prime muting is implemented as a scalar potential $V(x)$ on the Dirac operator ($D \to D + V$), then the Connes spectral distance is unchanged: $d(x, y) = \sup\{|f(x) - f(y)| : \|[D + V, f]\| \leq 1\} = \sup\{|f(x) - f(y)| : \|[D, f]\| \leq 1\}$, since $[V, f] = 0$ for multiplication operators. Consequently, $c_{\text{eff}} = c$.*

Inside the $G = 0$ bubble: flat Minkowski metric, standard light cone, no gravitational interaction with the exterior. The ship operates normally.

A. Spectral dimension of the Liouville shell

The spectral dimension $d_S(t) = -2\, d(\log K)/d(\log t)$ for $K_\pi(t) = \zeta(t)/\zeta(2t)$ at UV scales ($t \to 0$) gives $d_S \approx 0.4$—a drastic reduction from $d_S = 4$. At intermediate scales: $d_S \approx 2$.

This indicates *holographic dimension reduction*: the Liouville shell behaves as a 2D conformal surface. The bubble wall propagates according to 2D causal structure, which is not constrained by the 4D light cone.

The effective Virasoro central charge of the shell CFT is $c_{\text{Vir}} = 12\,\zeta(2)/\zeta(4) \approx 18$, indicating a non-trivial 2D conformal field theory.

VI. PHYSICAL IMPLEMENTATION

A. Prime-resonant quantum elements

Each prime p is addressed by a transmon qubit at frequency

$$\omega_p = \frac{\log p}{\tau}, \tag{15}$$

where τ is a reference timescale. For $\tau = 1\,\text{ns}$: $\omega_2 = 693\,\text{MHz}$, $\omega_3 = 1.10\,\text{GHz}$, $\omega_5 = 1.61\,\text{GHz}$—all standard superconducting qubit frequencies.

- **Absorption mode** ($\delta\epsilon_p = -1$): qubit absorbs the p-th Euler factor → muting. Two qubits → Theorem 2 ($G = 0$).
- **Phase mode** ($\phi = \pi$): qubit applies a Z-gate (π-phase shift) to the p-th sector → Liouville rotation. All qubits → Theorem 5 ($G = -2G$).

B. Combined two-layer bubble

Layer	Operation	G_{eff}	c_{eff}
Outer shell	π-rotate all	$-2G$	c ($d_S \approx 2$)
Inner region	Mute $\{2,3\}$	0	c ($d_S = 4$)
Exterior	None	G	c ($d_S = 4$)

VII. EXPERIMENTAL TEST

The regulator conjecture (that the zeta-regularized Casimir energy of a mode-filtered cavity equals the physical Casimir energy) makes a sharp prediction for a 2-prime-sieved acoustic cavity.

Setup. A $142\,\mu$m quartz BAW crystal with internal Mo/AlN Bragg reflectors at $L/3$ and $2L/3$ (implementing $p = 3$ muting) and DN boundary condition (implementing $p = 2$ muting).

Prediction.

$$\frac{|E_{\neg\{2,3\}}|}{|E_{\text{DD}}|} = \frac{|\zeta_{\neg\{2,3\}}(-3)|}{|\zeta(-3)|} = |(1-2^3)(1-3^3)| = 7\times 26 = 182. \tag{16}$$

A. Why 182 vs. 1/3: two regularizations

Both predictions are mathematically exact but arise from different regularization schemes applied to the same mode sum:

	Cutoff	Zeta
Evaluation point of $(1-p^{-s})$	$p^0 = 1$	p^3 ($s = -3$)
$p = 2$ factor	$1 - 1/2 = 1/2$	$1 - 8 = -7$
$p = 3$ factor	$1 - 1/3 = 2/3$	$1 - 27 = -26$
Product (ratio)	1/3	182

The cutoff regularization treats the mode sum as a smooth density ($\sum_{n\in\text{sieve}} n^3 / \sum_{\text{all}} n^3 \to 1/3$) and is "blind" to arithmetic structure. Zeta regularization preserves the Euler product through analytic continuation: each factor $(1 - p^{-s})$ is evaluated at $s = -3$, giving $(1 - p^3)$, which grows rapidly with p.

In standard (unmuted) Casimir physics, zeta regularization gives the experimentally verified result $E = -\pi^2\hbar c/(720d^3)$. However, the *ratio* between two different spectral geometries has not been tested for non-standard mode spectra. The 182-vs-1/3 experiment is the first such test.

Cost: \$2,200. **Timeline:** 4–6 months. A factor of 546 separates the two predictions—easily distinguishable if systematic errors are controlled to within one order of magnitude (temperature-sweep differential method).

B. Critical Casimir enhancement for the π-junction test

The critical Casimir effect [**?** **?**] amplifies vacuum forces near T_c by $(\xi/d)^{3/2}$ (mean-field universality class). For the Nb π-junction with $d = 2\,\text{nm}$ CuNi barrier and $\xi_0 = 38\,\text{nm}$:

T/T_c	ξ/d	Enhancement
0.99	190	3×10^3
0.9999	1900	8×10^4
0.999999	19000	3×10^6

This improves the $\Delta M/M$ ratio from $\sim 10^{-16}$ (far from T_c) to $\sim 10^{-9}$ (at $T/T_c = 0.999999$), well above the Eöt-Wash 10^{-13} precision. However, the *absolute* gravitational force remains below the detection threshold ($\sim 10^{-21}$ N vs. 10^{-14} N), because the nanoscale junction mass is intrinsically small.

The critical enhancement is most valuable for the *spectral* test (resonance frequency shift at T_c), not the gravitational test. Near T_c, the sign-reversal signal δf is amplified by $\sim 10^5$, making the π-phase Casimir sign change far easier to detect.

VIII. DISCUSSION

A. Relation to known no-go theorems

The Alcubierre metric requires weak energy condition (WEC) violation. Our framework does *not* violate the WEC: the $G = 0$ region has zero stress-energy coupling to curvature, and the $G = -2G$ region is repulsive but involves positive energy (phase-rotated modes, not exotic matter). The metric distortion arises from the *spectral* structure, not from the stress-energy tensor.

B. Assumptions and limitations

The central assumption is that the BC spectral triple $(\mathcal{A}, \mathcal{H}, D)$ with $D^2 = H_{\mathrm{BC}}$ faithfully describes the physical vacuum. This is testable (Section VII).

Additional caveats:

1. The BC system is $(0+1)$-dimensional (no native spatial structure). Embedding in $3+1$ dimensions requires a product geometry $M \times \mathrm{Spec}(\mathbb{Z})$ that is not fully formalized.

2. The Seeley-DeWitt expansion at $t = 0$ may receive non-perturbative corrections not captured by the polynomial expansion.

3. The Liouville heat kernel $\zeta(t)/\zeta(2t)$ has a pole at $t = 1/2$ (from $\zeta(2t = 1)$); the physical meaning of this singularity is unexplored.

C. Entropy cost

The Landauer bound for maintaining the muted state is $W \geq k_B T|\Delta S|$ per mode. At 10 mK: $W \sim 10^{-8}\,\mathrm{J\,m^{-3}}$—negligible compared to the metamaterial wall energy ($\sim 10^3$ J).

IX. CONCLUSION

We have shown that the Euler product structure of the Riemann zeta function, interpreted as a decomposition of the BC vacuum into independent prime channels, permits two forms of gravitational coupling control: nullification ($G = 0$, Theorem 2) and reversal ($G = -2G$, Theorem 5).

Both operations are achieved by manipulating individual prime sectors of the Euler product—muting (absorption) for $G = 0$, and π-rotation (Z-gate) for $G < 0$—using standard circuit QED technology.

The $G = 0$ interior preserves $c_{\mathrm{eff}} = c$ (Connes distance proof), while the $G = -2G$ shell undergoes spectral dimension reduction to $d_S \approx 2$ (holographic), providing a potential FTL mechanism via 2D causal structure.

A single experiment—measuring the Casimir energy ratio 182:1 in a 2-prime-sieved BAW cavity ($2,200)—falsifies or confirms the physical realizability of both theorems, constituting the most cost-effective test of arithmetic vacuum engineering.

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